Supervised Learning Recap

Kianté Brantley
CS 1840 Fall 2026

Announcements

HW3 is out and due 10/29!

Midterm will be in 1 week on Mon 10/26!

Check the course website for Office Hours:
https://harvard-cs-1840.github.io/2026/

Finishing Up From Last Lecture

✓ **1. Simulation lemma:**

What is the performance of π under any estimator $\widehat{P}$

✓ 2. **Algorithm:** estimate $(\widehat{P}, \widehat{r})$ from data and compute $\widehat{\pi}^\star$—the optimal policy of $(\widehat{P}, \widehat{r})$

3. **Analyzing** the performance $\widehat{\pi}^\star$ under (P, r)

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$$\widehat{P}(s' \mid s, a) = \frac{\sum_{i=1}^{N} \mathbf{1}\{s_i' = s'\}}{N};$$

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2. Planning w/ the learned model:

$$\widehat{\pi}^\star = \textbf{PI}\left(\widehat{P}, r\right)$$

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w/ probability at least, $1 - \delta$ we will have $|\hat{p} - p| \leq O\left(\sqrt{\frac{\ln(1/\delta)}{N}}\right)$

(concentration bound; Hoeffding's inequality; proof out of scope)

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We can show:

$$\|\hat{P}(\cdot \,|\, s, a) - P(\cdot \,|\, s, a)\|_1 \lesssim \sqrt{1/N}, \forall s, a$$

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Performance of the learned policy

Now assume that $\hat{P}$ is nearly accurate, i.e., $\|\hat{P}(\cdot|s,a) - P(\cdot|s,a)\|_1 \leq \epsilon, \forall s, a$

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(Simulation lemma)

$$\leq \frac{\gamma}{(1-\gamma)^2}\left[\mathbb{E}_{s,a\sim d_{s_0}^{\pi^\star}}\|\widehat{P}(\cdot|s,a) - P(\cdot|s,a)\|_1 + \mathbb{E}_{s,a\sim d_{s_0}^{\hat{\pi}^\star}}\|\widehat{P}(\cdot|s,a) - P(\cdot|s,a)\|_1\right]$$

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$$\le \frac{2\gamma}{(1-\gamma)^2}\cdot\epsilon,$$

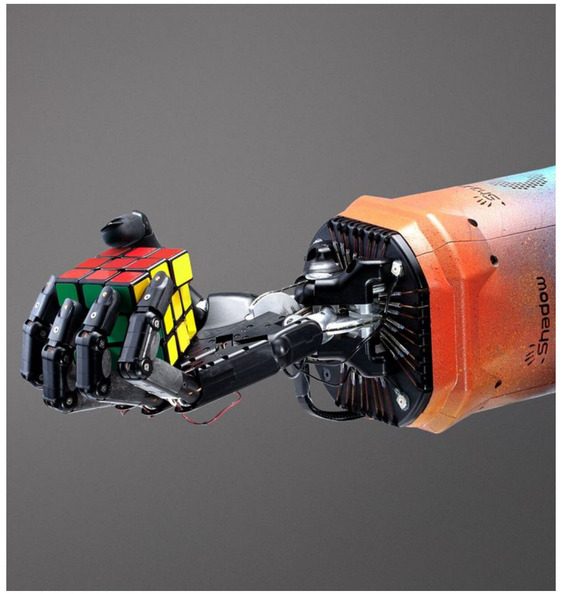
Recap:

So far, three learning algorithms:
TD learning, Q learning, and model-based RL

Limitation: they only work for small MDPs with discrete states and actions

Recap:

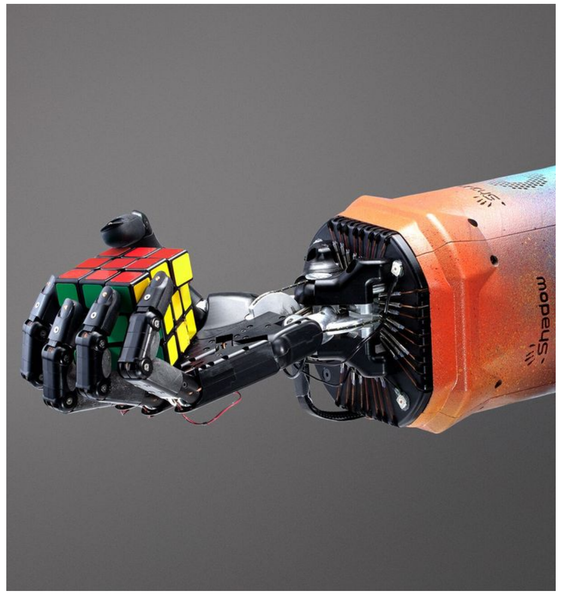
Real world problems often have continuous state or extremely large number of states





Recap:

Real world problems often have continuous state or extremely large number of states





Cannot hope to **enumerate** all possible state-actions in reality…

Starting from today:

Making RL work for large-scale MDPs with the help from supervised learning
(e.g, Deep Learning)

Outline

Recap supervised learning

Tutorial on PyTorch and Gym

Multi-class classification

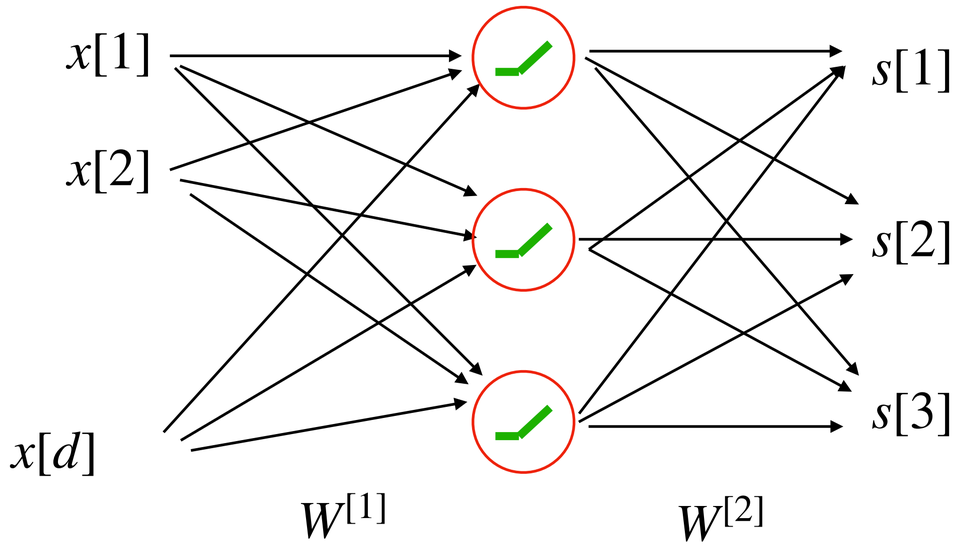
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Goal: learn the distribution over labels $P(\cdot \,|\, x)$

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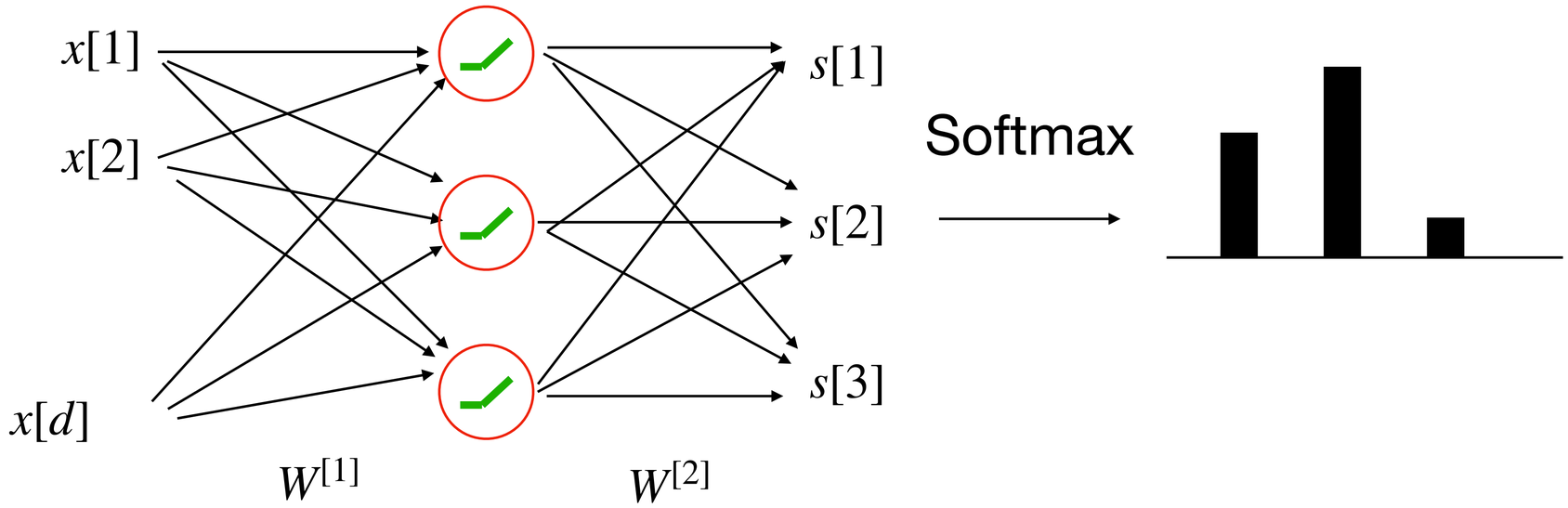
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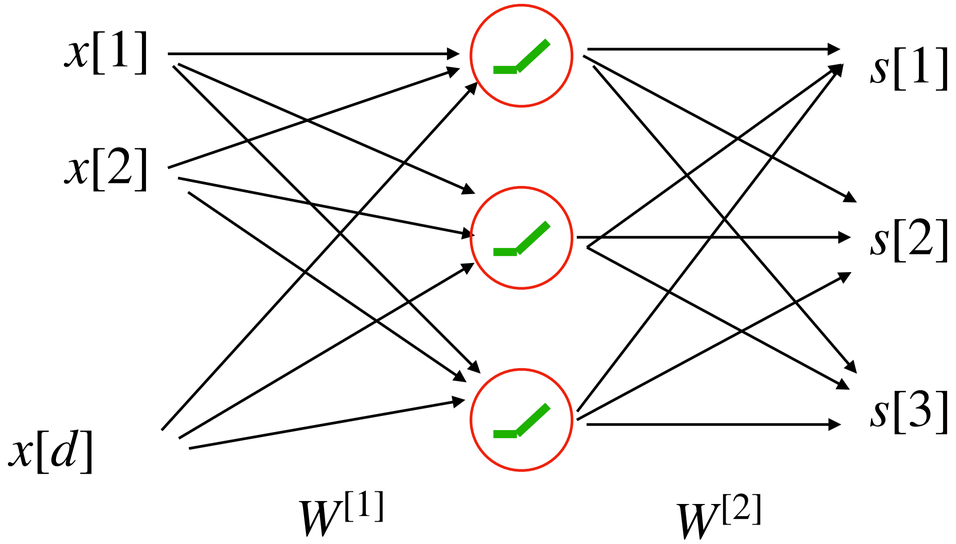
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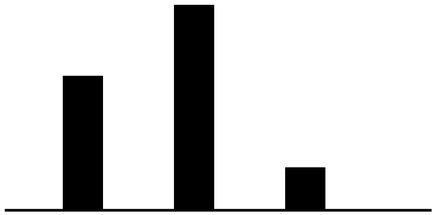
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Softmax

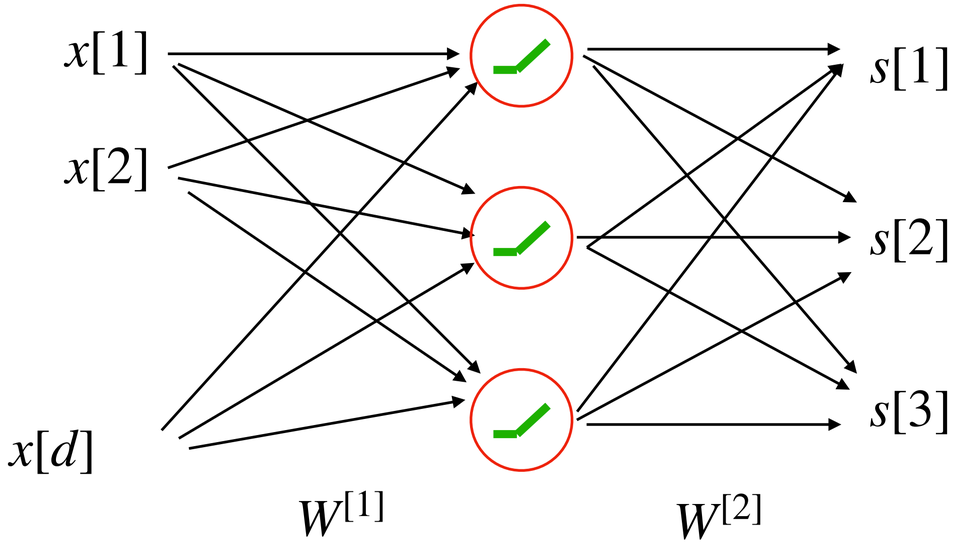


$$P(y = i \mid x) = \frac{\exp(s[i])}{\sum_{j=1}^{K} \exp(s[j])}$$

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Softmax



$$P(y = i \mid x) = \frac{\exp(s[i])}{\sum_{j=1}^{K} \exp(s[j])}$$

$$P_\theta(\cdot \mid x) = \text{softmax}\left(W^{[2]}\text{ReLu}(W^{[1]}x)\right)$$

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Negative log-likelihood

$$\ell(\theta) = \frac{1}{N} \sum_{i=1}^{N} -\ln P_\theta(y^i \mid x^i)$$

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Maximize the likelihood of labels given features

Regression

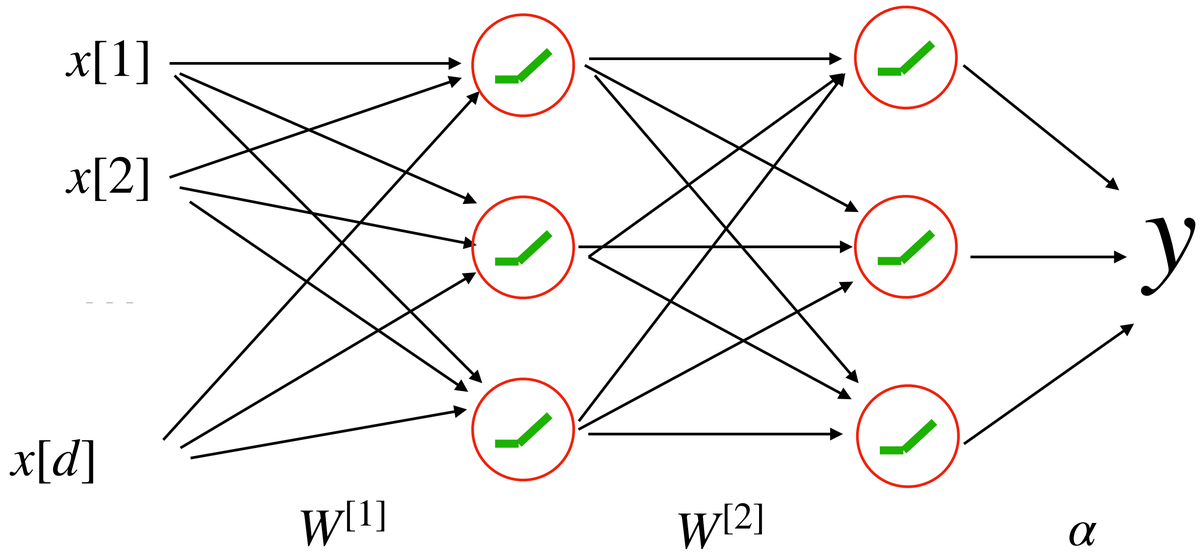
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Goal: learn the **Bayes optimal** $\mathbb{E}[y\,|\,x]$

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Minimize the mean squared error

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Dist where training data is sampled

Generalization

Supervised learning exhibits generalization ability, as long as test samples are sampled from the same training dist

e.g., Classifer trained on large-scale cat / dog images can classifier **unseen** cat or dog images

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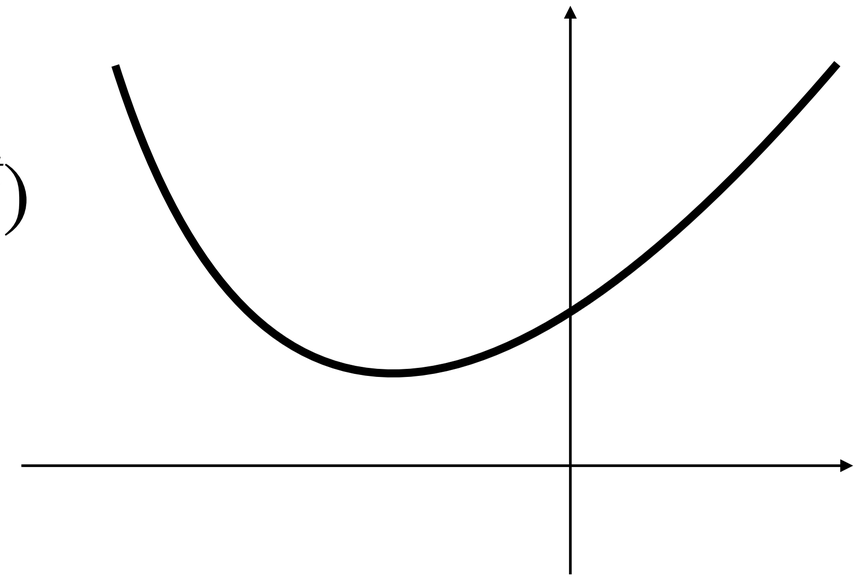
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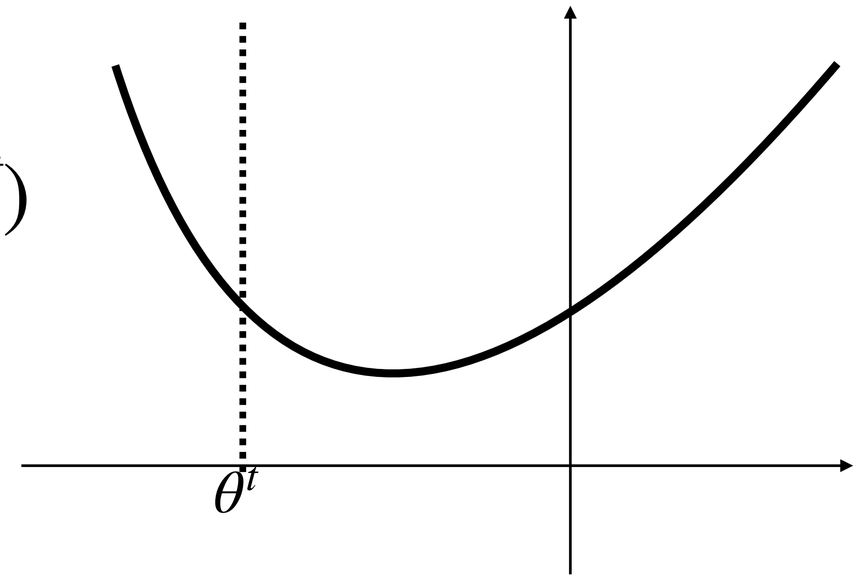


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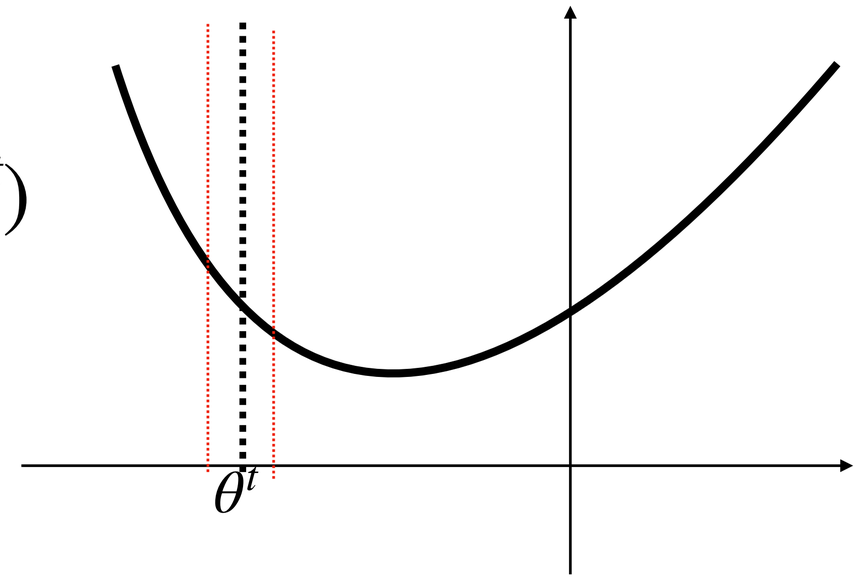


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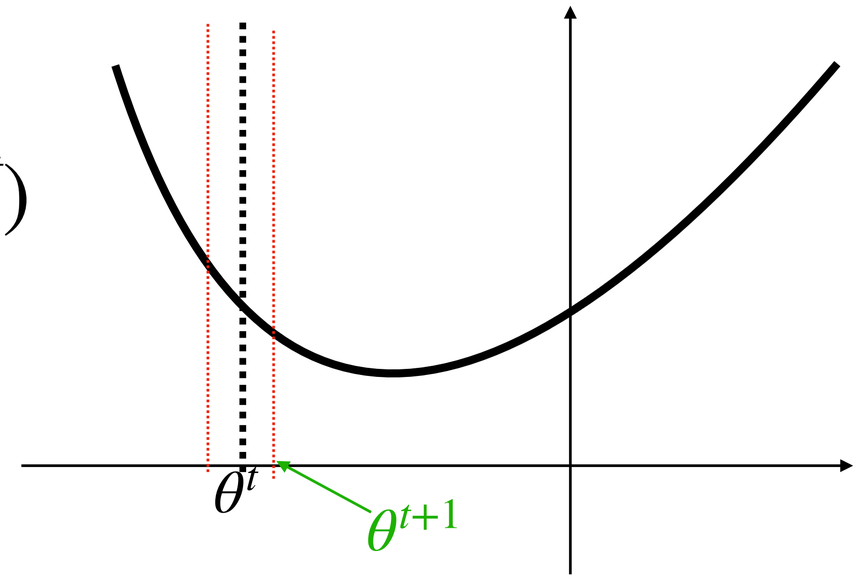


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Q: how to get this unbiased estimate of the gradient?

Optimization

Often we use adaptive gradient methods such as Adagrad or Adam:

In high level, adaptively set learning rates for different coordinates and time

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Feedback and Attendance

https://bit.ly/4gYn2nC

